

1 Introduction

The Blue-Eyed Islanders riddle, also known as the Muddy children riddle (Fagin et al., 1995), is a textbook example illustrating how agents reason from public information and higher-order knowledge. After seeing examples it becomes relatively intuitive how the riddle unfolds, explaining its widespread use to exhibit public announcement logic.

Within dynamic epistemic logic, public announcement logic (PAL) focuses on the effects of (truthful) public announcements on agents’ knowledge (Van Ditmarsch & Ruan et al., 2007). These announcements are available to all agents and restrict agents to consider worlds solely in line with the announced information. In other words, perfect logical reasoning will enable agents to gain new knowledge from these announcements. Additionally, agents may update their knowledge from expected behaviour (or inaction) of other agents following an announcement. PAL allows analyses of how agents, from common knowledge, epistemically reason about the knowledge and states of other agents based on the observed actions.

The original Blue-Eyed Islanders riddle introduces the following scenario: consider an island inhabited by a group of people. Out of this group, k people have blue eyes while the rest have brown eyes. Everyone is able to see the eye colour of the other inhabitants. Nonetheless, nobody knows their own eye colour. All of this is common knowledge for all islanders. One day, an outsider arrives at the island and makes the following announcement: “At least one of you has blue eyes.”. The island has a special rule that all the inhabitants must follow: if someone discovers they have blue eyes, they are obligated to leave the island at the next dawn. Therefore, the only information the islanders receive is the announcement and the fact whether islanders leave or not. It is common knowledge that the inhabitants never lie nor cheat and are perfect logicians.

From extensive validations and implementations (Gierasimczuk & Szymanik, 2011; Macovei, 2023; Truřaş et al., 2023; Van Benthem, 2003, 2007), it has become clear that for k blue-eyed islanders, it will be the k th dawn when all blue-eyed islanders leave. The reasoning and conclusions are well-understood and will also be explained later in this work. However, one common assumption in the puzzle is the fact that all islanders can see each other (except themselves). This visibility assumption provides that any original Blue-Eyed Islanders riddle with n agents can be expressed as an $n \times n$ matrix of binary digits.

An entry 1 in such a matrix indicates that the column agent can see the eye colour of the row agent, while 0 indicates that they cannot. For any $n \times n$ binary matrix, there ought to be $2^{n \cdot n}$ different possible configurations and $2^{n \cdot (n-1)}$ if the diagonal is fixed to 0. Consequently, we can revisit the classical riddle from a different point of view. The role of vision and the different implementations in various puzzles is an interesting topic in the field (Balbiani et al., 2012; Gasquet & Schwarzenruber, 2011; Schwarzenruber, 2009). To elaborate, we are interested to what extent the assumption of complete visibility is necessary for the islanders to find out whether they have blue eyes. The question that we want to answer is: “For which exact visibility matrices will all blue-eyed islanders eventually know their own eye colour within a finite number of days?”. This question grasps the purpose of our revisit, namely an attempt to generalize the implementations and conclusions to the Blue-Eyed Islanders riddle for different visibility configurations. This implementation will thus require translating asymmetric visibility matrices into Kripke models with the purpose of solving the puzzle. To answer the question, we assume standard S5 epistemic semantics and public announcement logic (Van Ditmarsch & Van Der Hoek et al., 2007).

On this website, you can expect the details of our implemented models and state updates and the theoretical background for the implementation. Moreover, the methods of implementations and restrictions will be reported together with the findings and conclusions. For illustrative

purposes, there will be a demo available where you can try out our implementation yourself for any configuration you desire. This is all done with the purpose of answering our proposed question and therefore to provide more generally applicable insights on this well-known riddle.